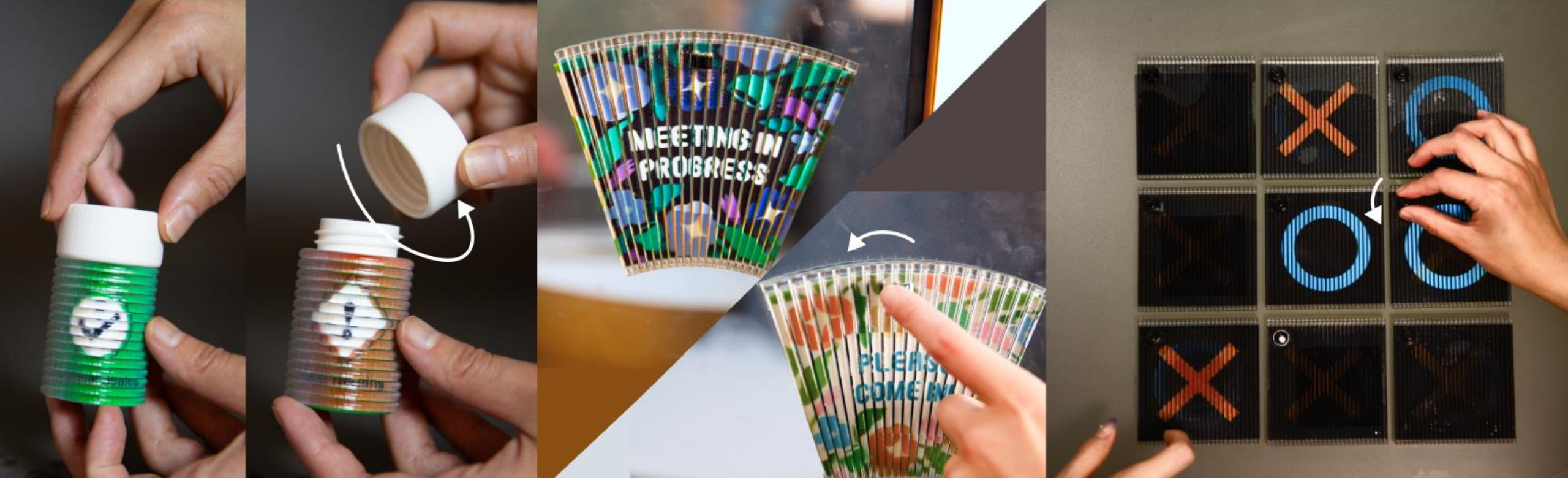
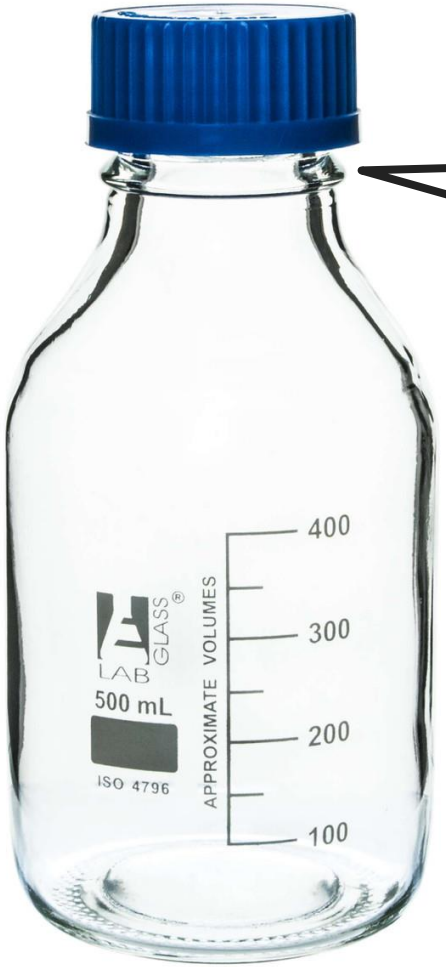



# ShiftLens

## Mechanically Actuated Optical Surfaces for Switchable Appearances on 3D Objects

Yunyi Zhu, Dingning Cao, Jeremy Mrzyglocki, Stefanie Mueller, Narges Pourjafarian

# Hidden States in Everyday Objects




Is this bottle tightly closed?

Is this light on?



# Electronic State Indicators

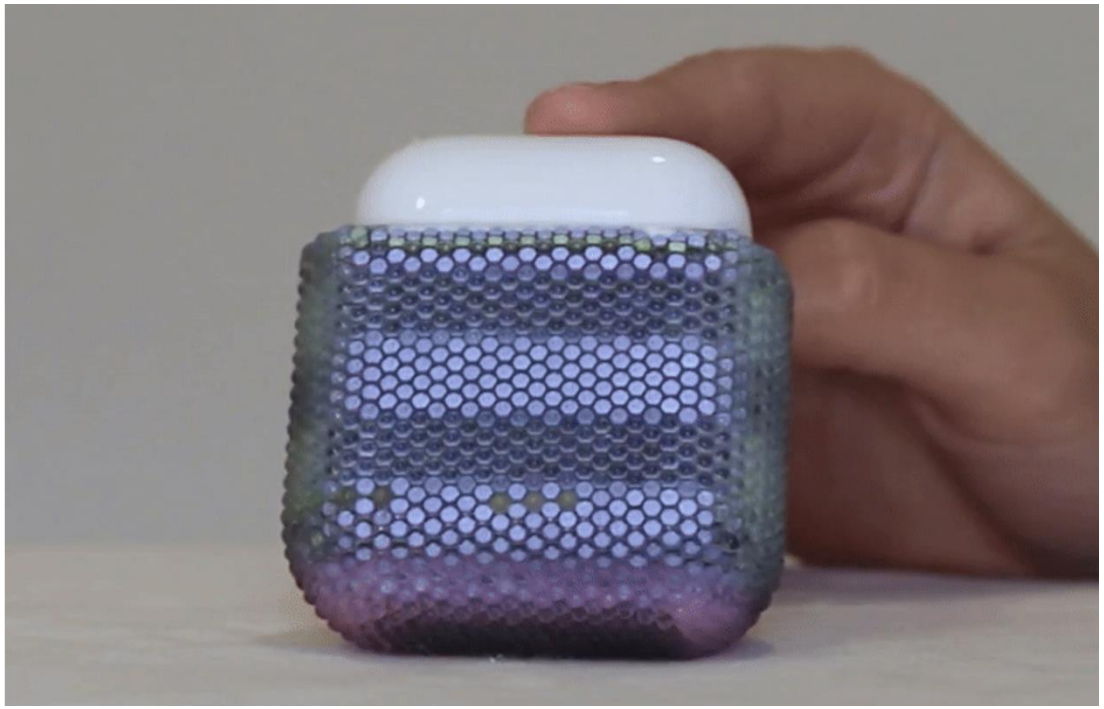




OUR APPROACH:

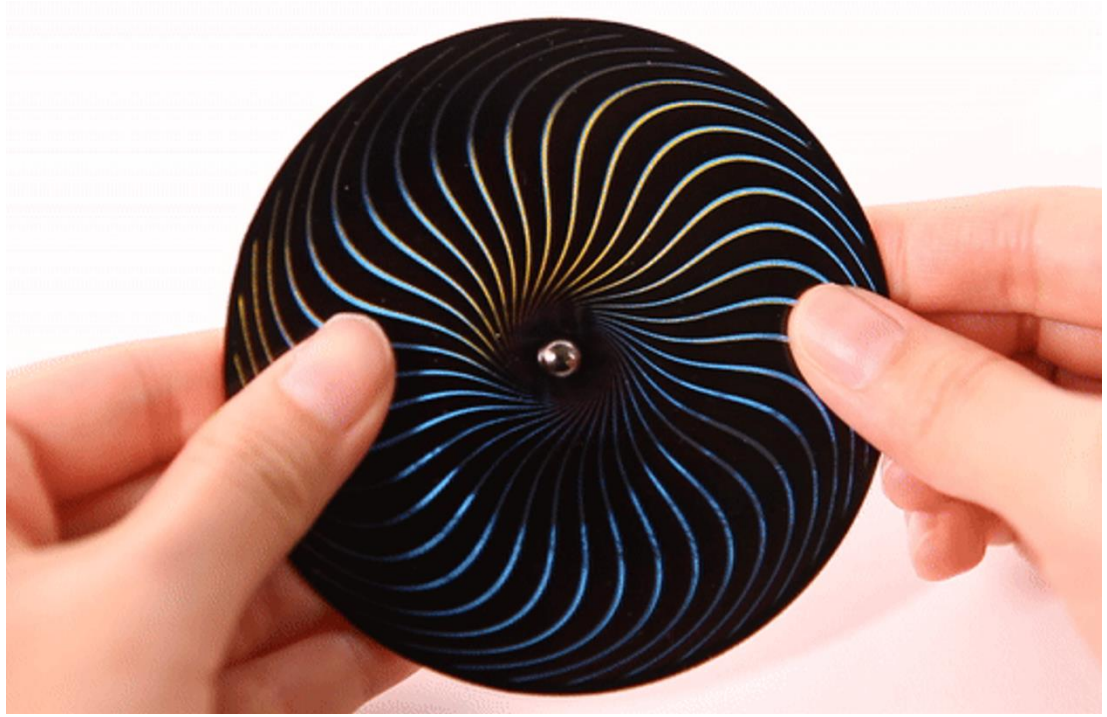
# Surface Optics

# Related Work



## Lenticular Lenses

**Lenticular Objects** *(UIST 2021)*



## Barrier-grid Animation

**FabObscura** *(UIST 2025)*



# Lenticular Lenses

**Lenticular Objects** *(UIST 2021)*

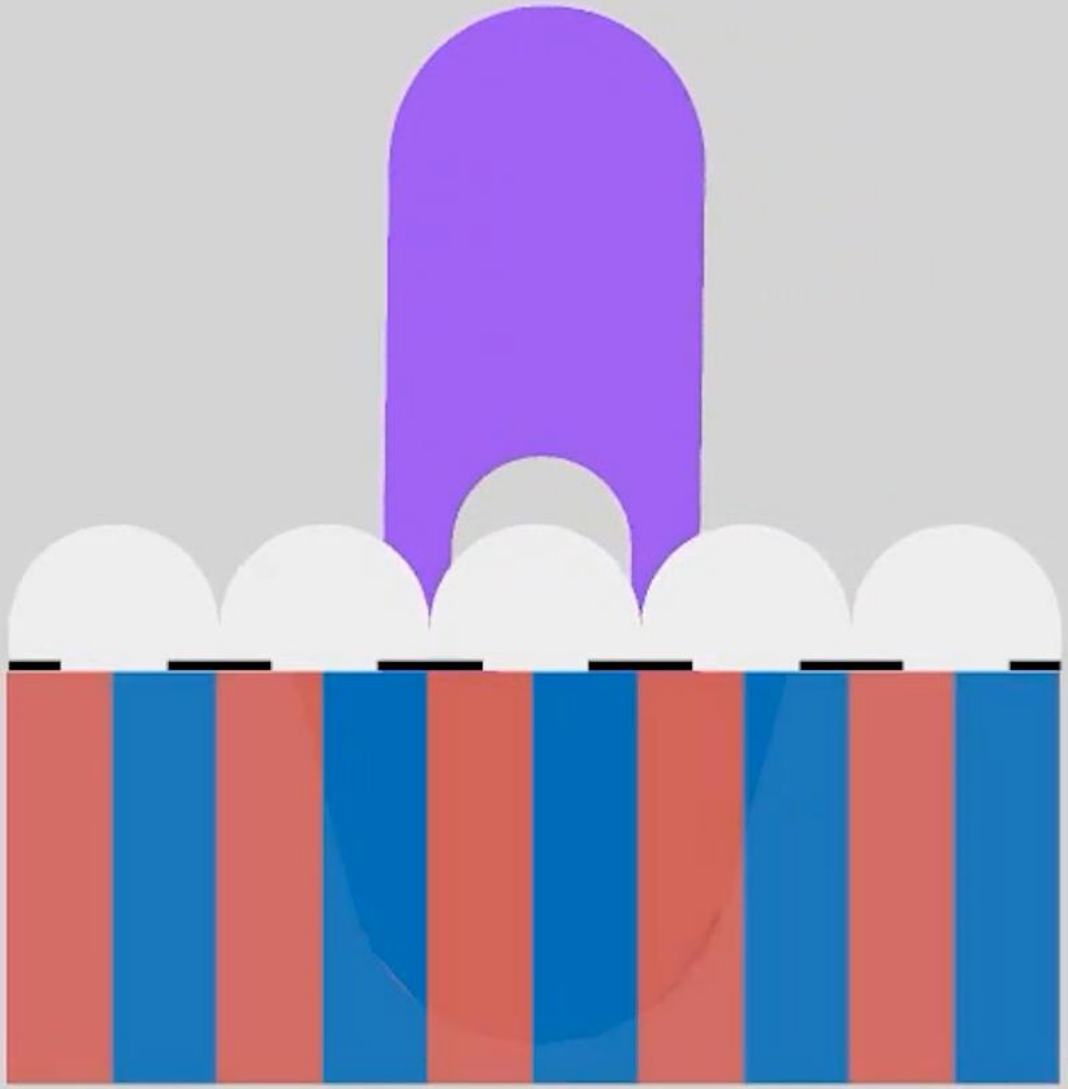


# Barrier-grid Animation

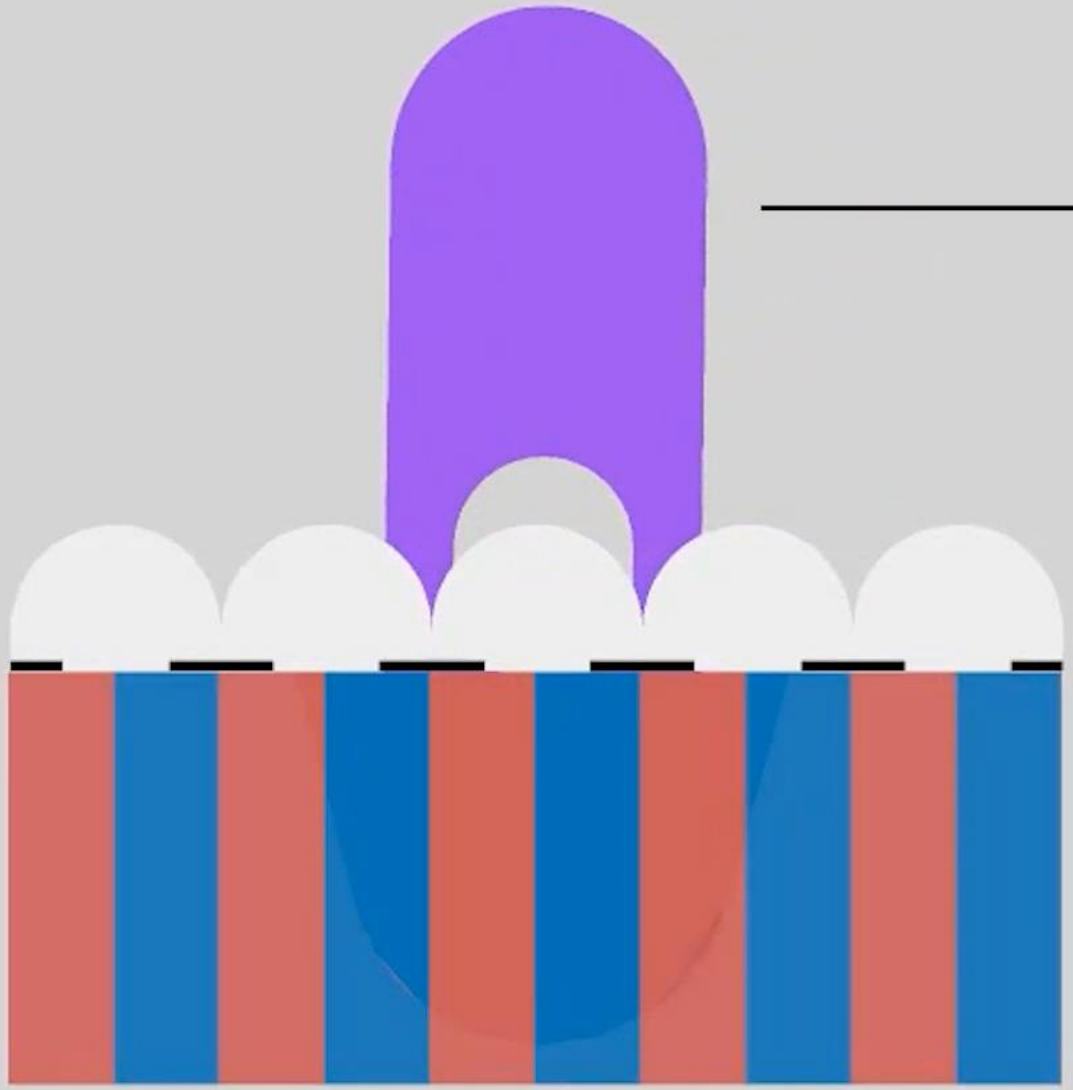
**FabObscura** *(UIST 2025)*

# ShiftLens





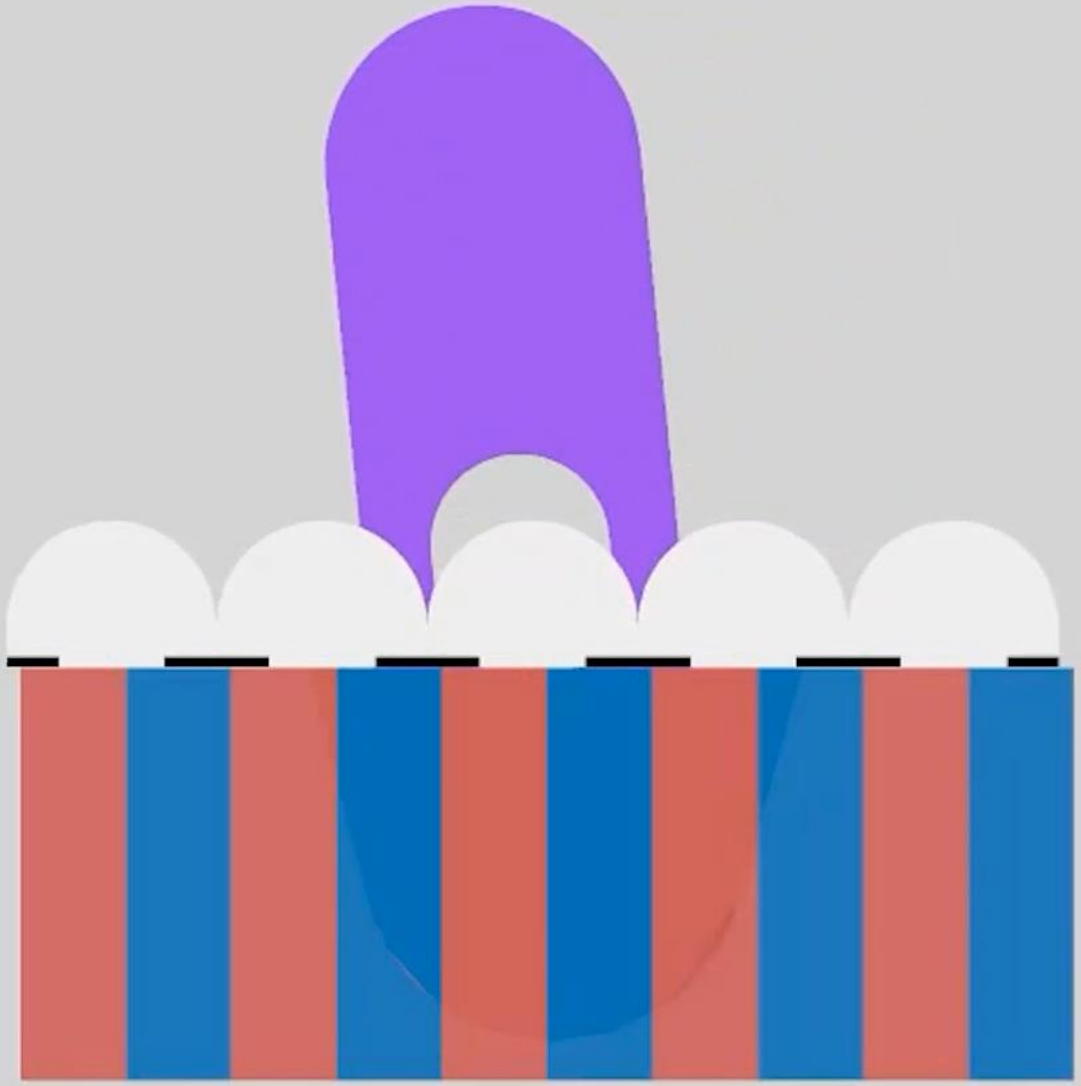
ShiftLens Structure

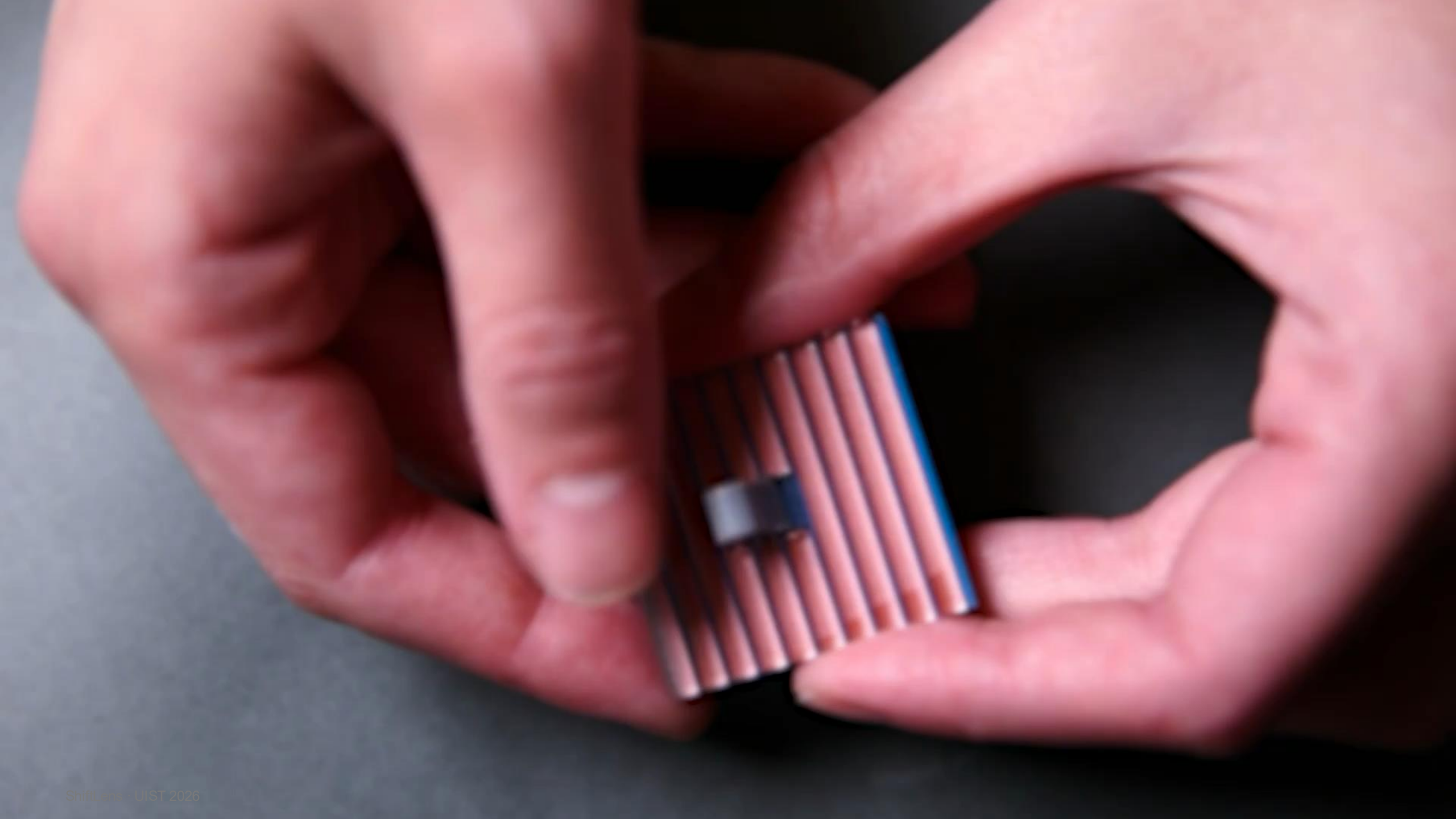


Actuation Mechanism

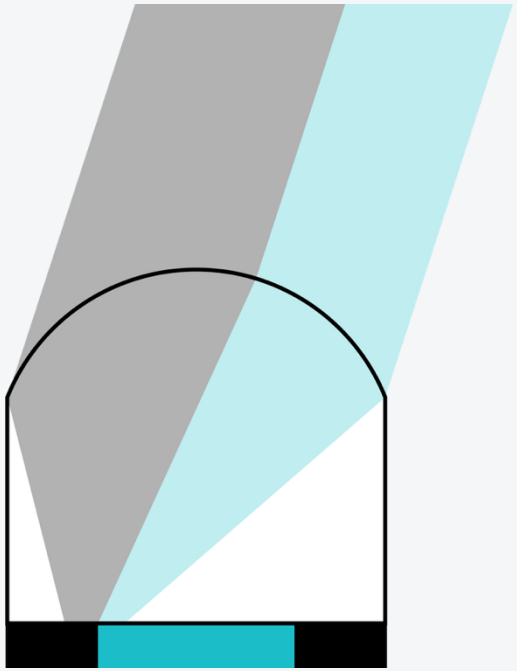
Lens Layer

Pattern Layer

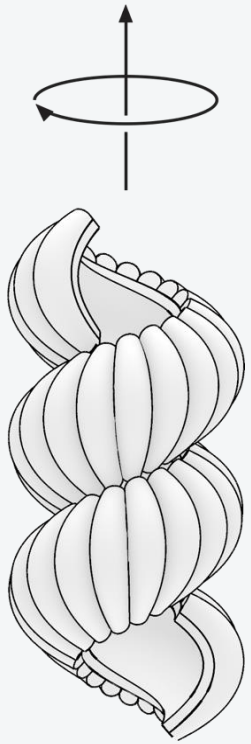




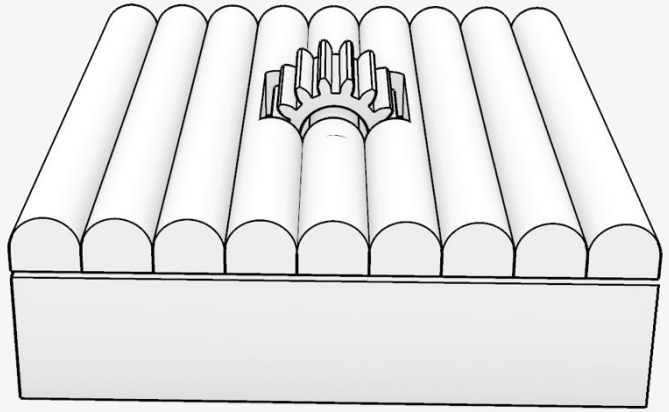
# Technical Challenges



## 1 Optical

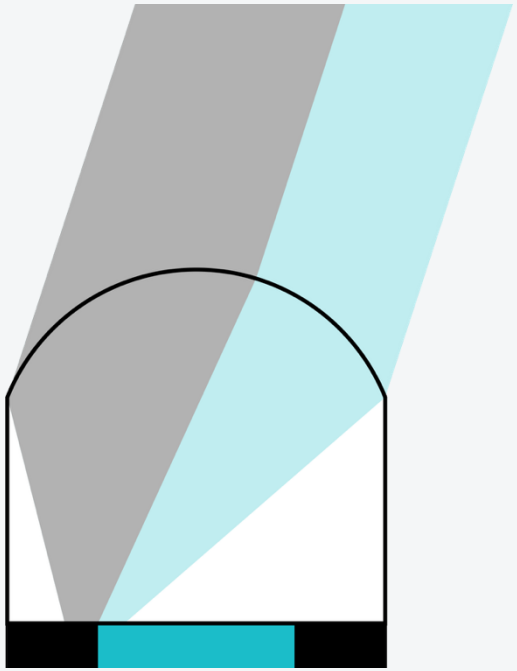


## 2 Geometric



## 3 Mechanical

# Technical Challenges



## 1 Optical

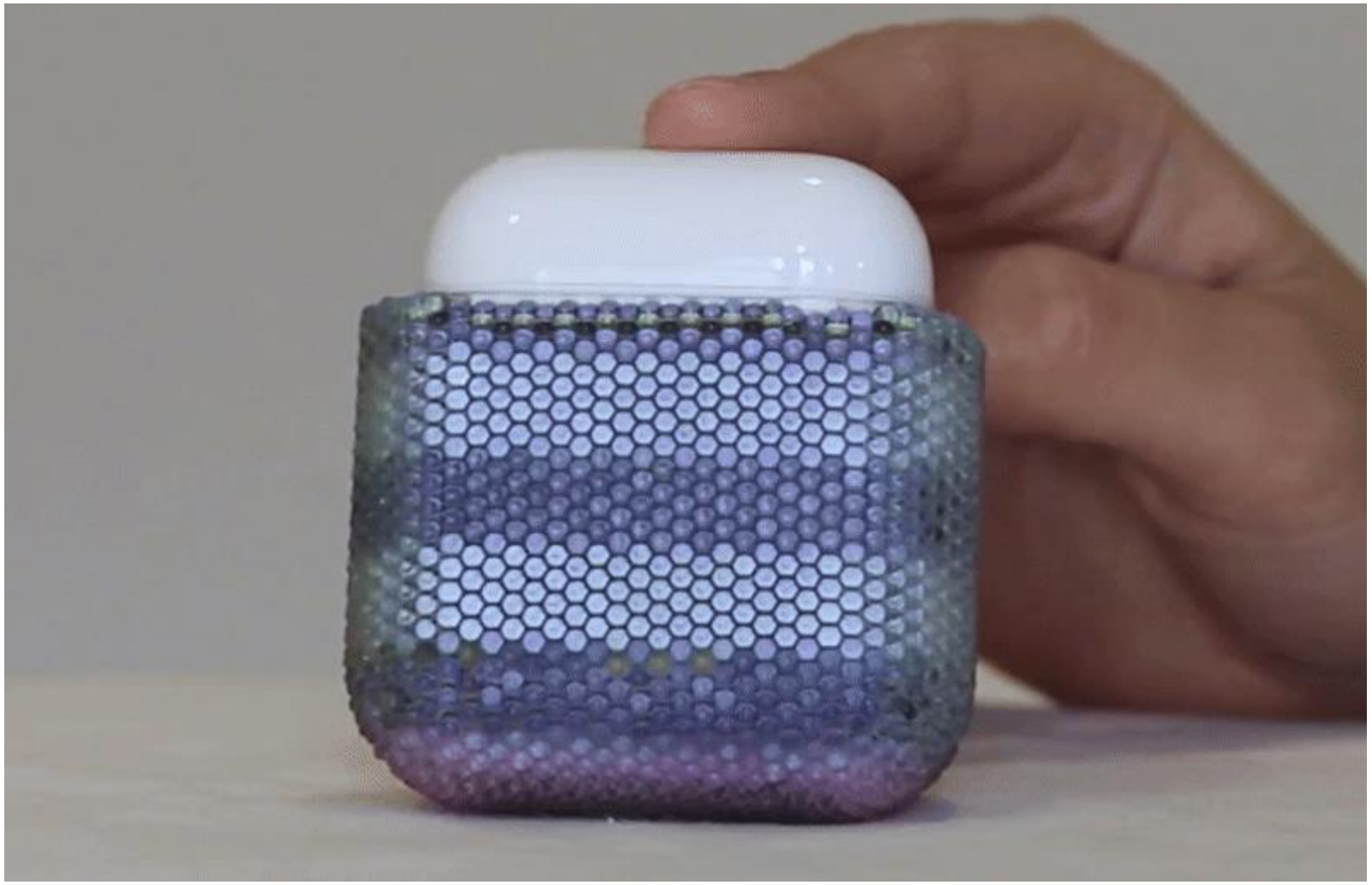


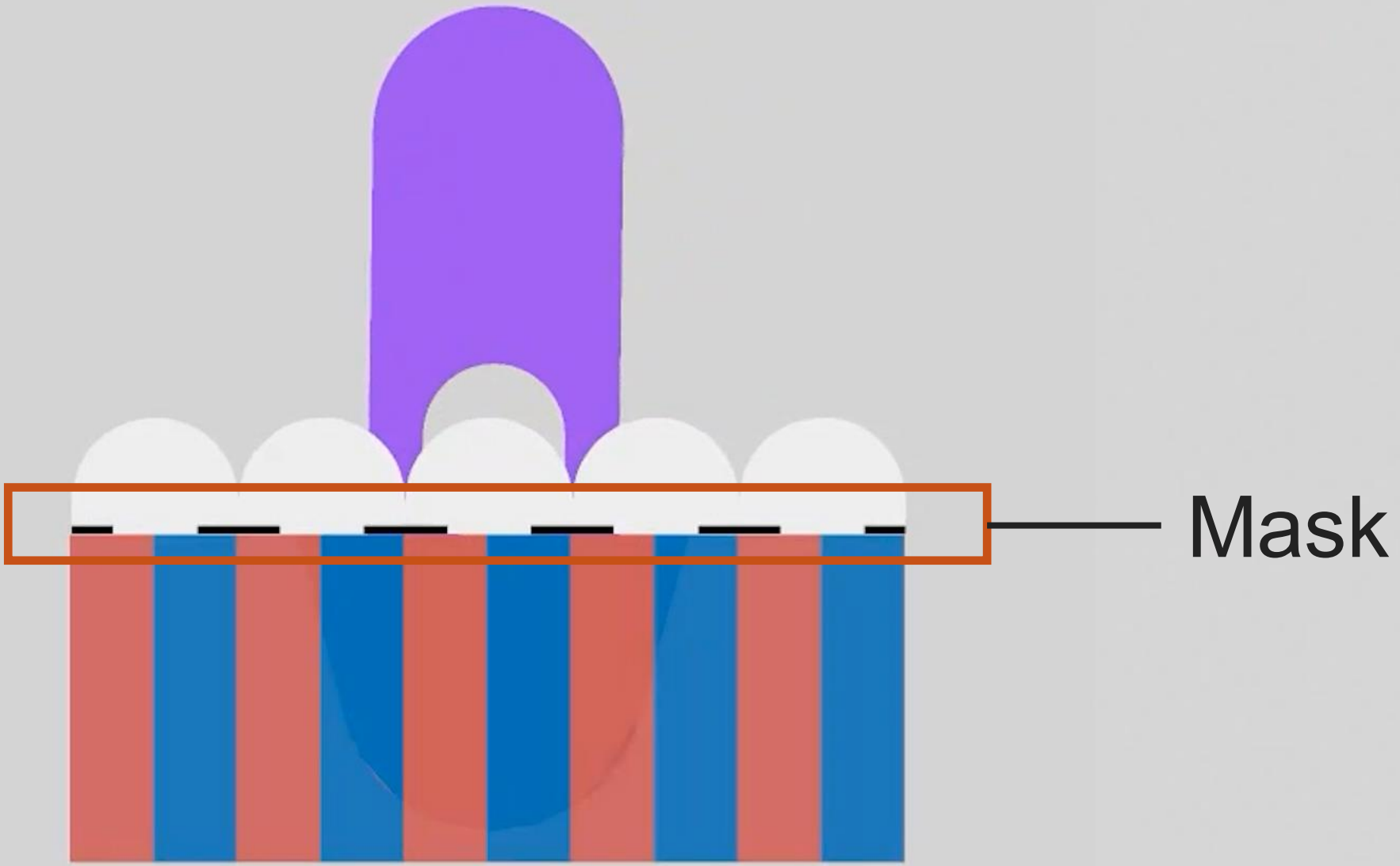
## 2 Geometric



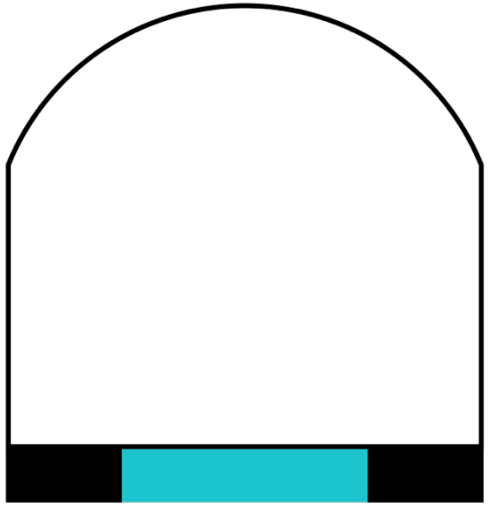
## 3 Mechanical

# Optical Challenge: View Dependence



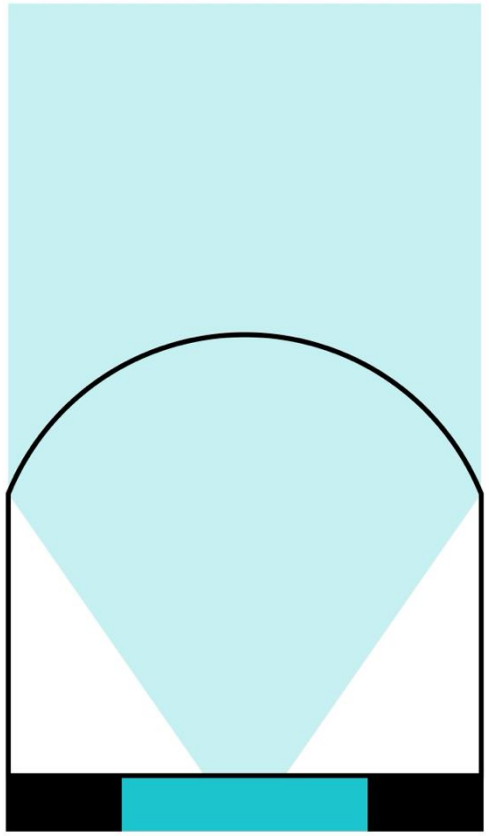

Mask

# Lens Geometry Optimization



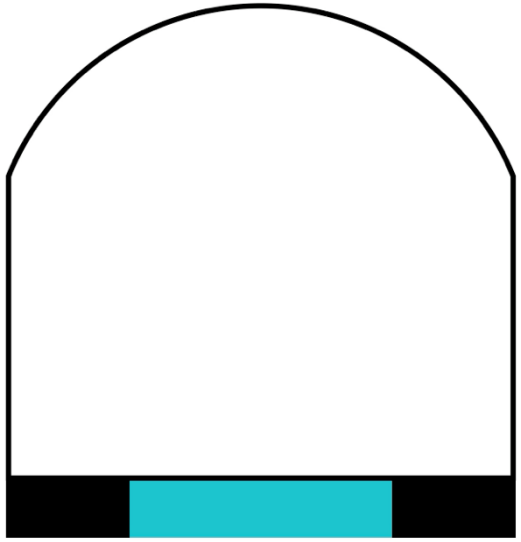
Conventional Lenticular Lens

# Lens Geometry Optimization



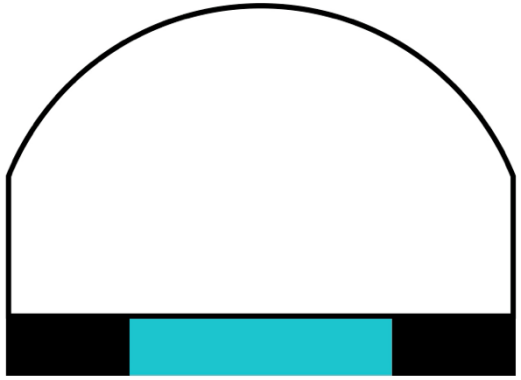
Conventional Lenticular Lens

# Lens Geometry Optimization



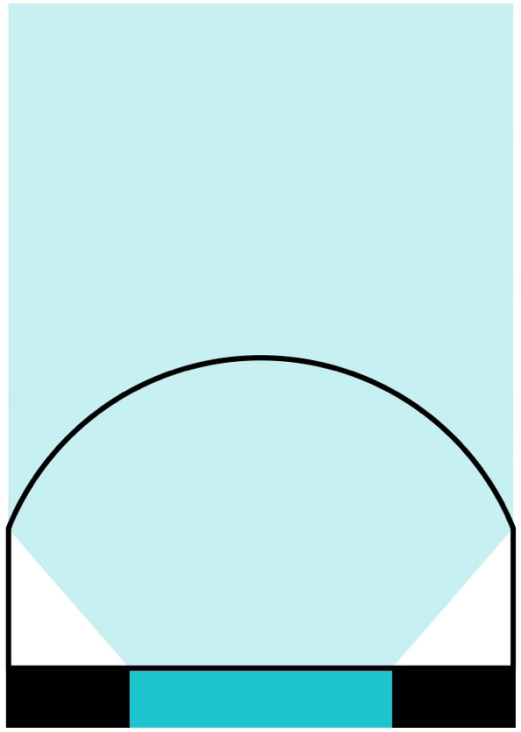
ShiftLens

# Lens Geometry Optimization



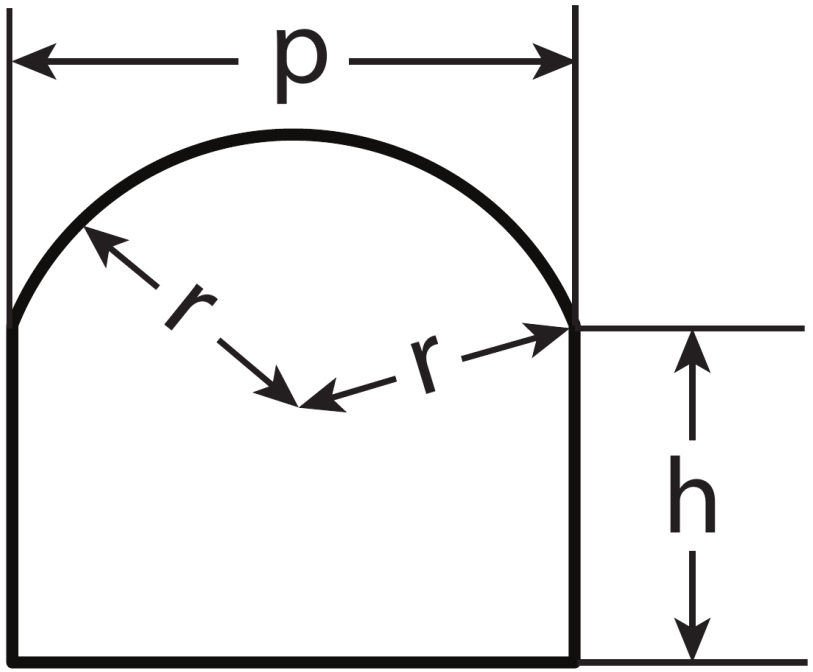
ShiftLens

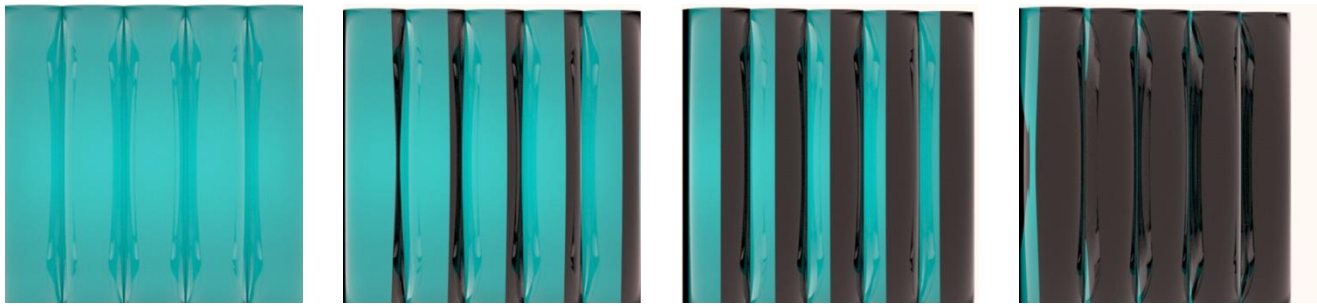
# Lens Geometry Optimization



ShiftLens

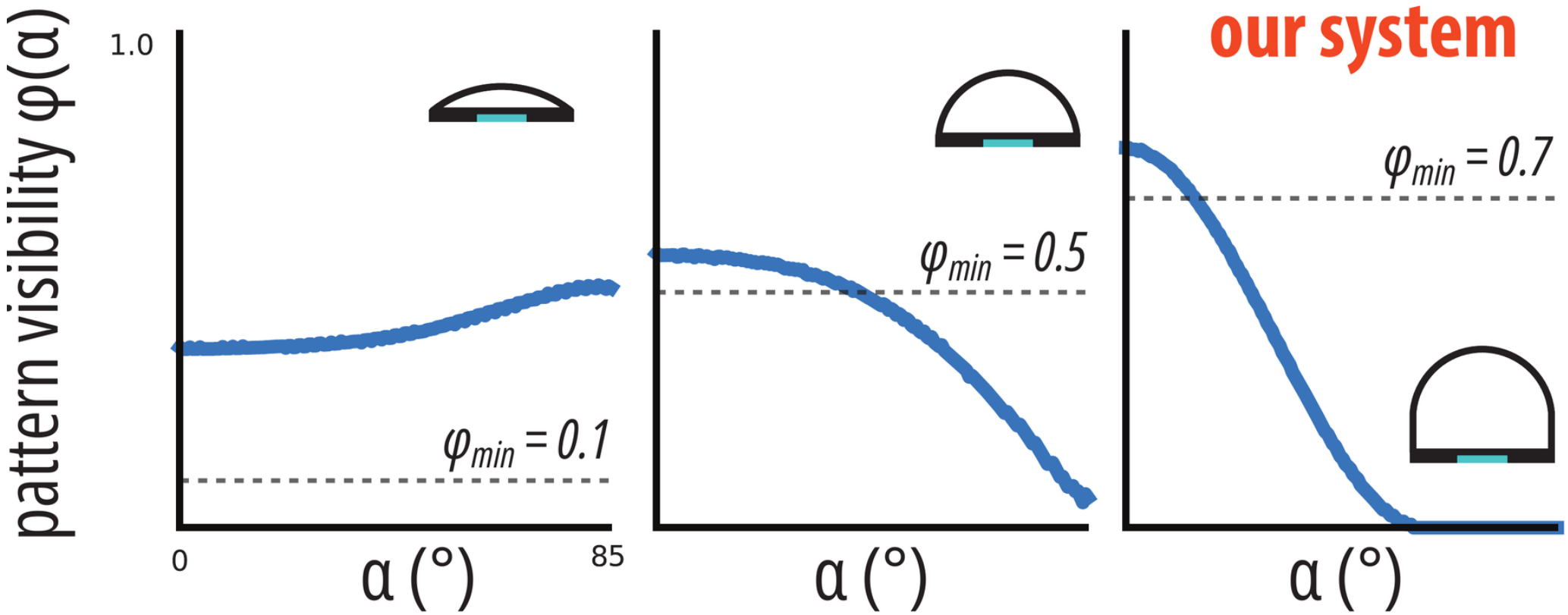
# Pattern Visibility



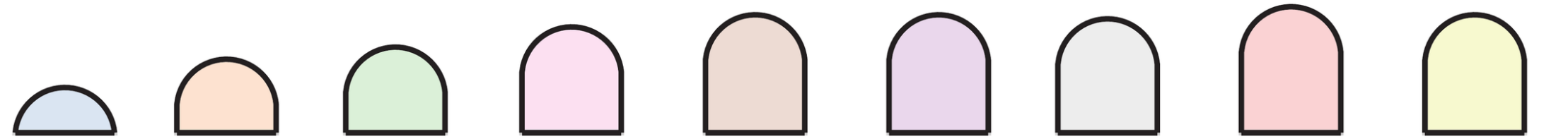



| 0° | 10° | 20° | 30° |
|---|---|---|---|
| φ = 1 | φ = 0.88 | φ = 0.34 | φ = 0 |

# Pattern Visibility vs. Viewing Range



# Optimized Lens Geometry

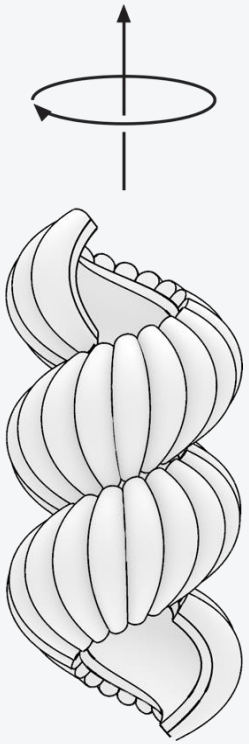


| N = 2 | N = 3 | N = 4 | N = 5 | N = 6 | N = 7 | N = 8 | N = 9 | N = 10 |
|---|---|---|---|---|---|---|---|---|
| r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p | r = 0.5p |
| h = 0 | h = 0.3p | h = 0.65p | h = 0.78p | h = 0.73p | h = 0.73p | h = 0.78p | h = 0.82p | h = 0.82p |
| θ = 85° | θ = 51° | θ = 26° | θ = 28° | θ = 25° | θ = 24° | θ = 25° | θ = 26° | θ = 26° |

# Technical Challenges



1 Optical

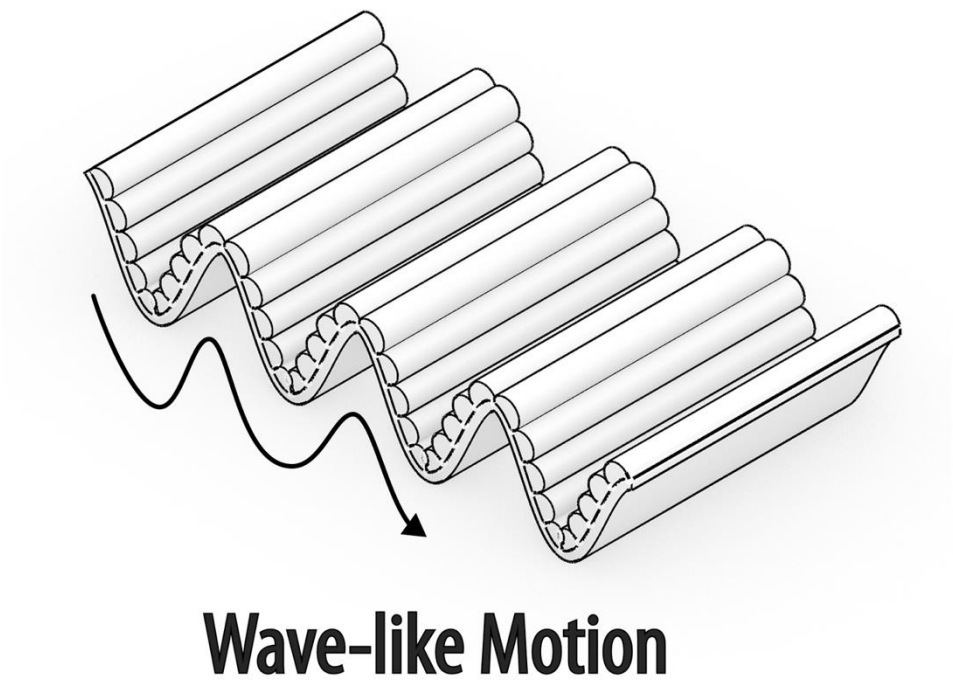


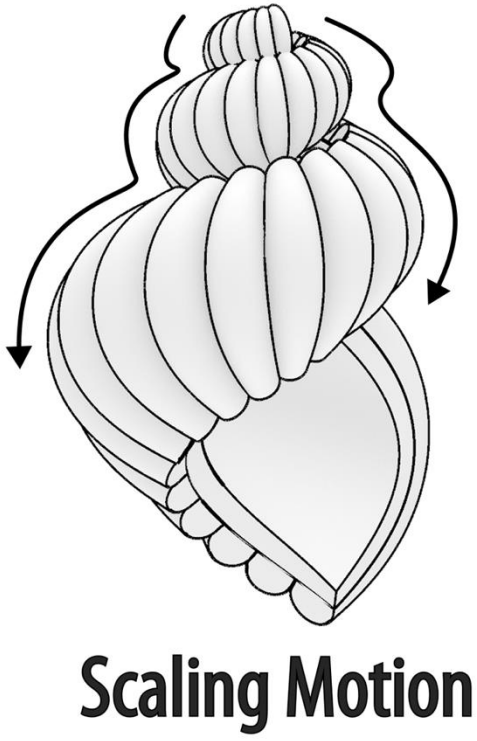
2 Geometric



3 Mechanical

# Incompatible Surfaces



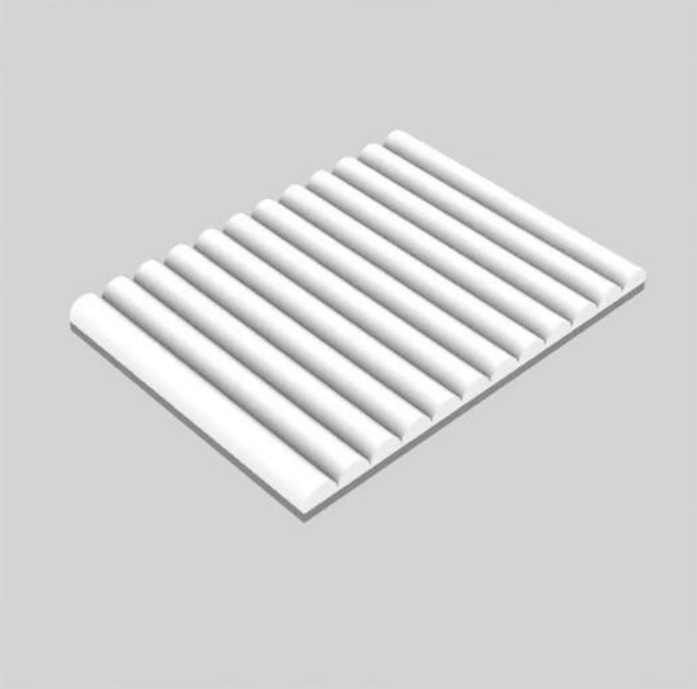



Chasles' Theorem:

# Any rigid motion is a screw motion

the most general displacement (or Euclidean motion) of a rigid body can be represented as a screw displacement: a combination of a rotation about an axis and a translation parallel to that same axis

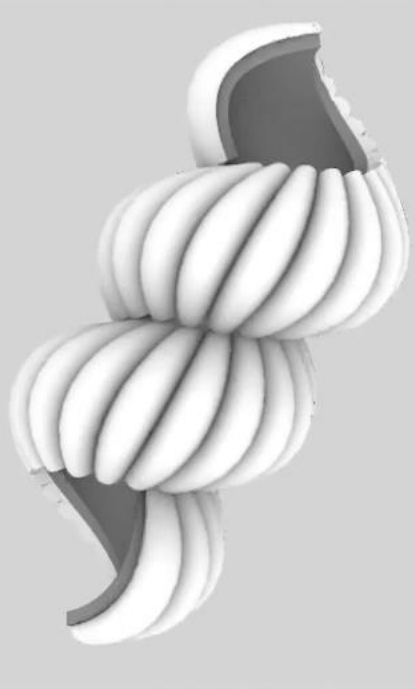
# Slidable Surfaces



Translation



Revolution



Combined

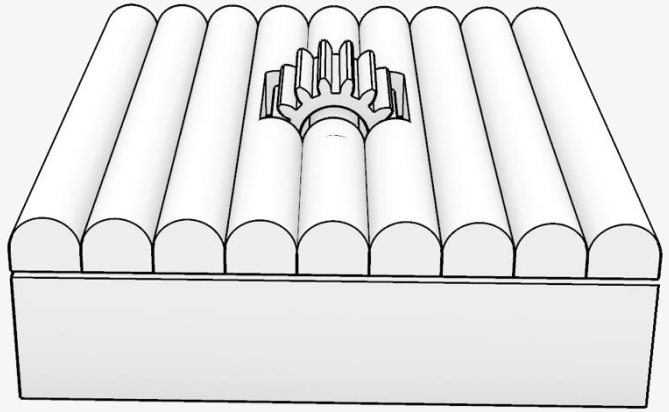
# Technical Challenges



1 Optical

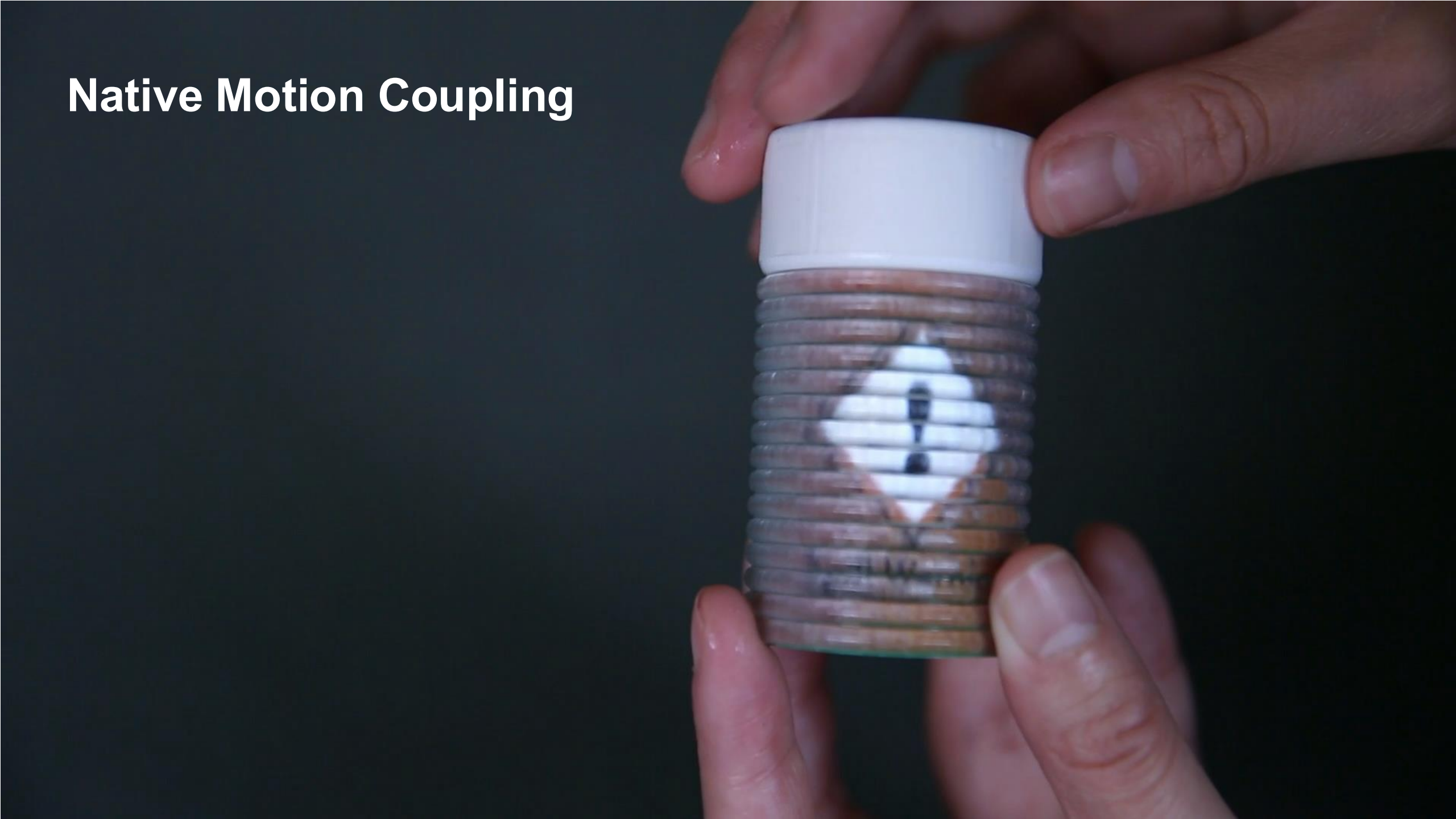


2 Geometric

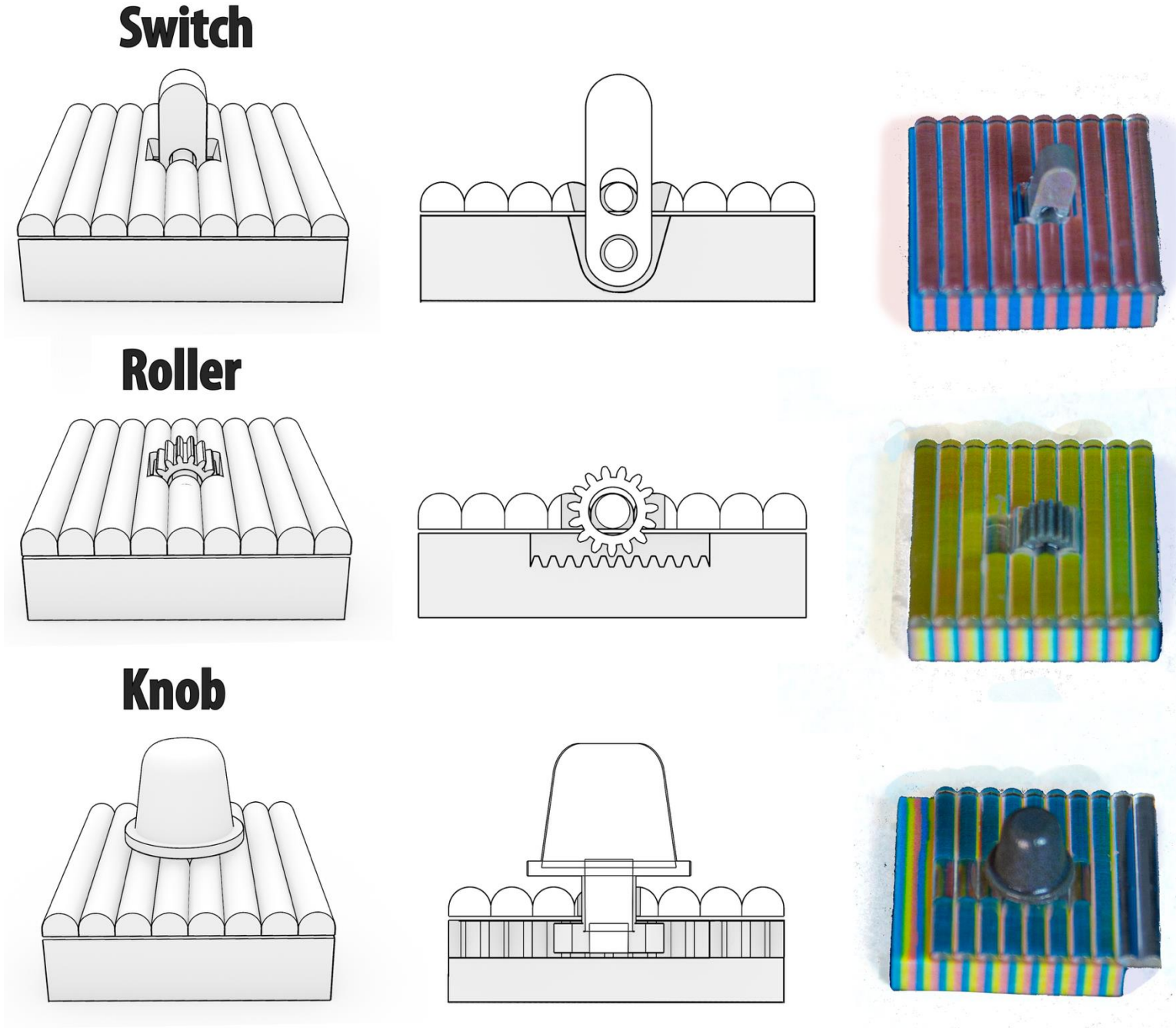


3 Mechanical

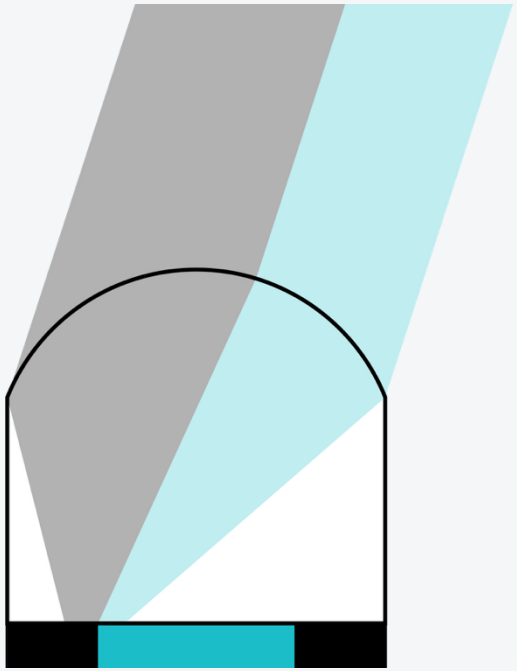
# Native Motion Coupling



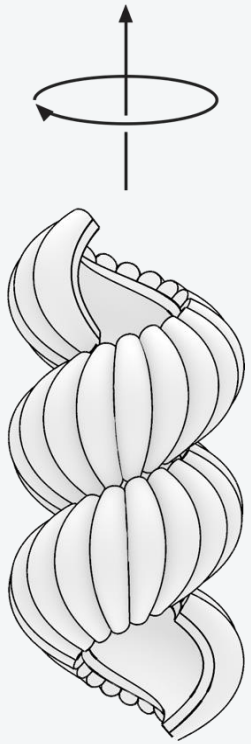
# Actuation Mechanisms



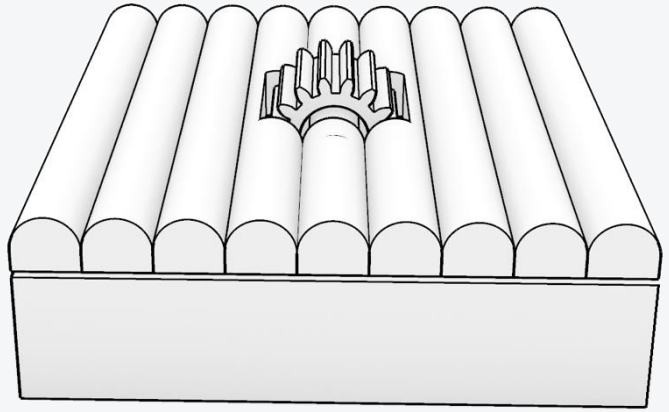
# Technical Challenges



## 1 Optical



## 2 Geometric



## 3 Mechanical

ShiftLens Design Tool

## Step 1: Define Surfaces and Lenses

**Choose Surface Type**



**Select Primary Curve** Select

**Select Revolve Axis** Select

**Revolve Angle** 40

**Lens Width** 5

**Flip Lens Side** Flip

**Add Rail** 3

**Surface Thickness** 3

## Step 2: Actuation Mechanism and Images

**Select (Optional) Actuation Mechanism**

User Custom

Add a State

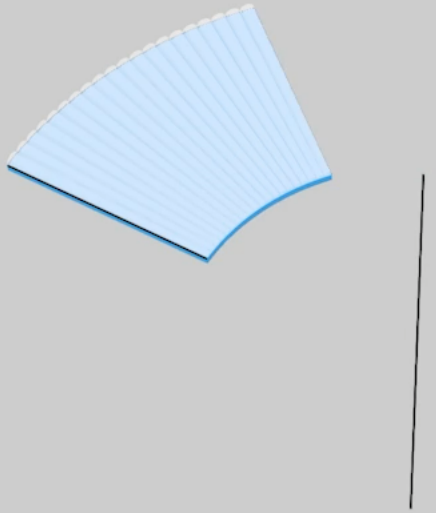
## Step 3: Simulate and Export

**Start Simulation** Start

**Export Files** Export

Command

Perspective



ShiftLens Design Tool

## Step 1: Define Surfaces and Lenses

**Choose Surface Type**



**Select Primary Curve** Select

**Select Revolve Axis** Select

**Revolve Angle** 40

**Lens Width** 5

**Flip Lens Side** Flip

**Add Rail** 3

**Surface Thickness** 3

## Step 2: Actuation Mechanism and Images

**Select (Optional) Actuation Mechanism**

User Custom

Add a State

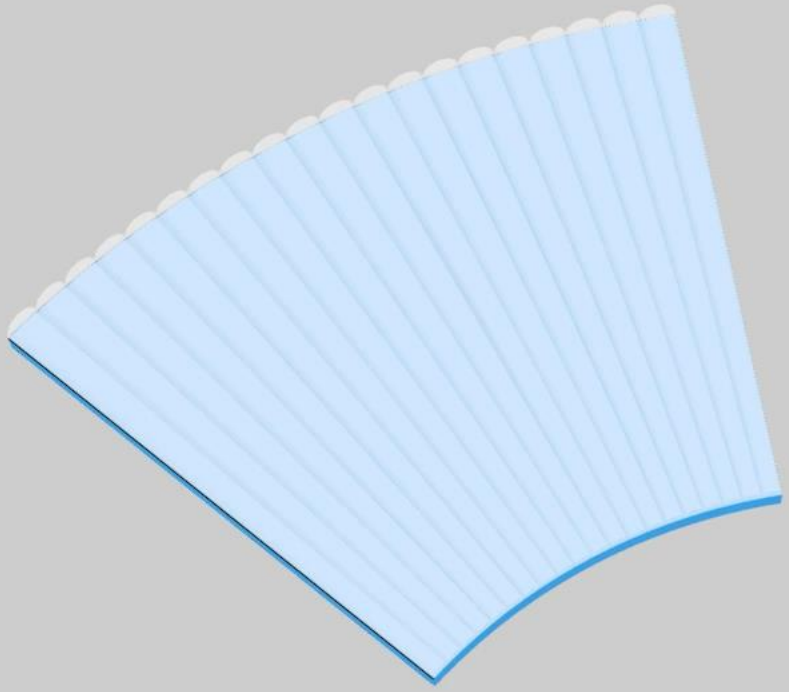
## Step 3: Simulate and Export

**Start Simulation** Start

**Export Files** Export

Command

Perspective



ShiftLens Design Tool

## Step 1: Define Surfaces and Lenses

**Choose Surface Type**



**Select Primary Curve** Select

**Select Revolve Axis** Select

**Revolve Angle** 40

**Lens Width** 5

**Flip Lens Side** Flip

**Add Rail** ☑ 3

**Surface Thickness** 5

## Step 2: Actuation Mechanism and Images

**Select (Optional) Actuation Mechanism**

Switch

Place the Switch

**Upload Images at Different States**

*Switch State 1:*

Browse...

Preview State in Viewport

*Switch State 2:*

**Pick switch placement — green=OK red=too close to edge (Esc=cancel):**

Point

Cancel

Done

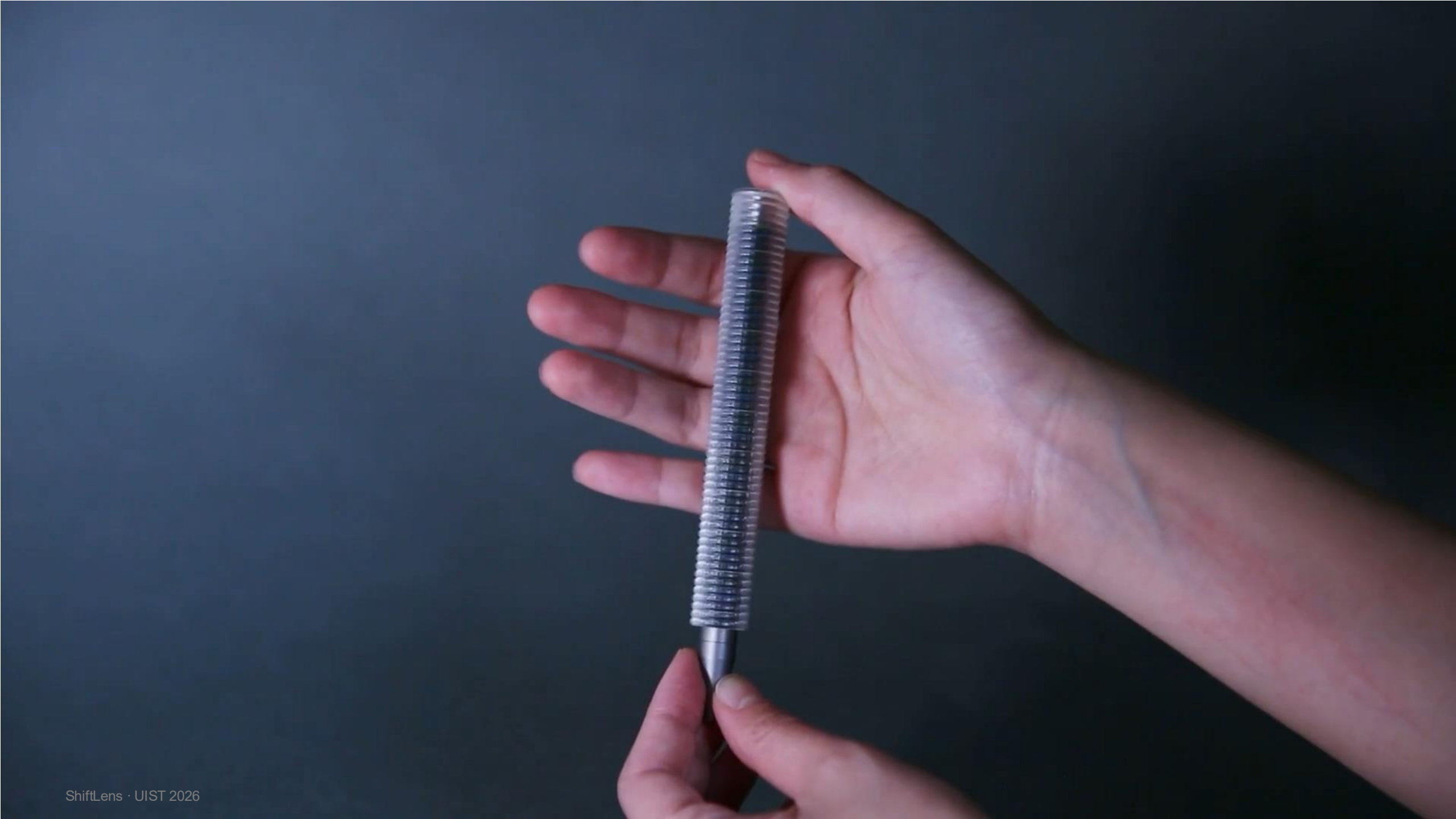
Perspective

OnSrf

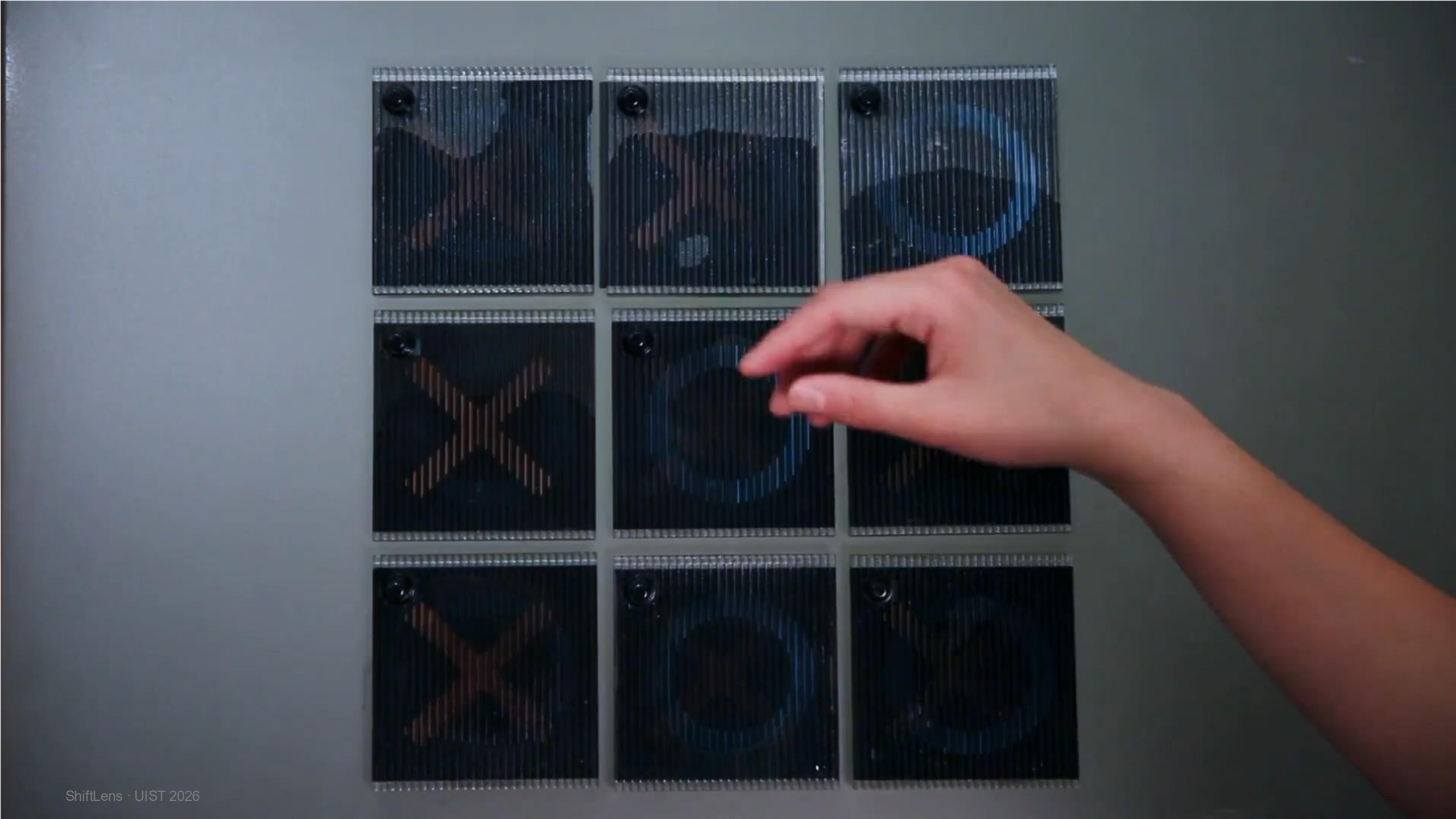

PLEASE
COME IN

# Applications

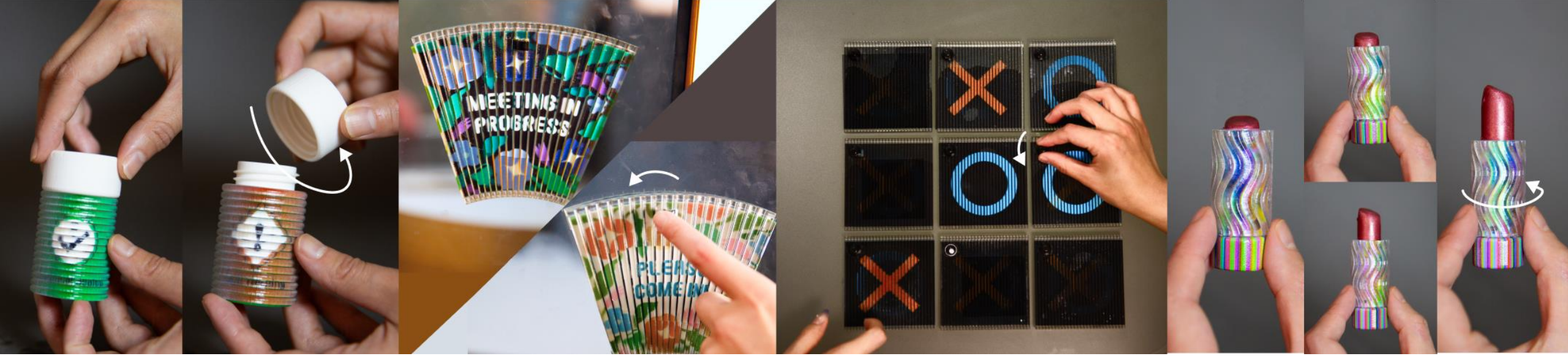





framery








# ShiftLens

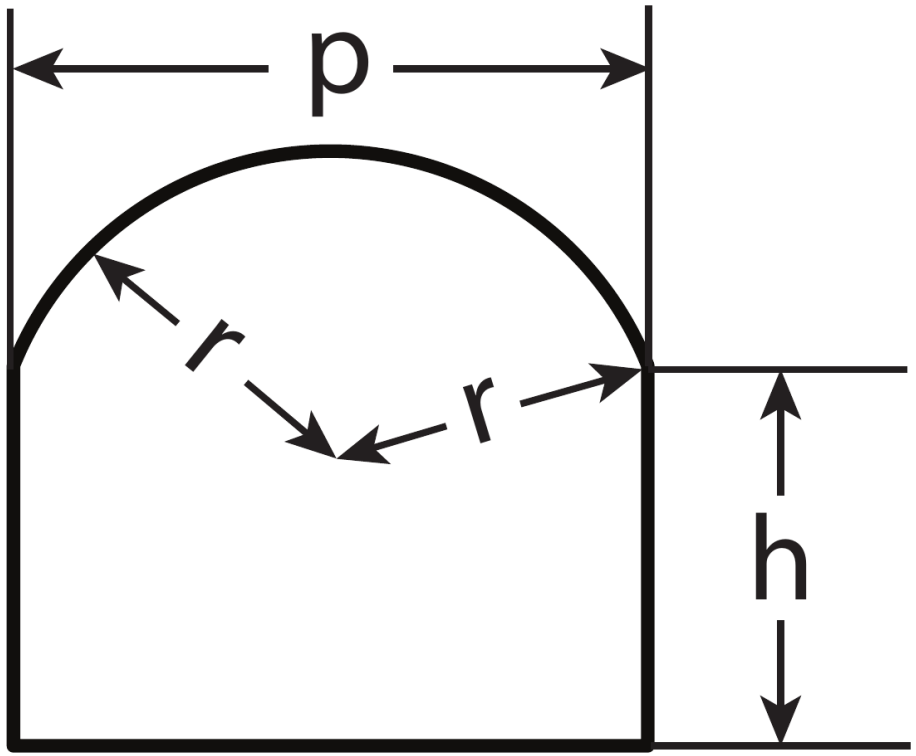
## Mechanically Actuated Optical Surfaces for Switchable Appearances on 3D Objects

Yunyi Zhu, Dingning Cao, Jeremy Mrzyglocki,
Stefanie Mueller, Narges Pourjafarian

Project Page

Open Source Design Tool

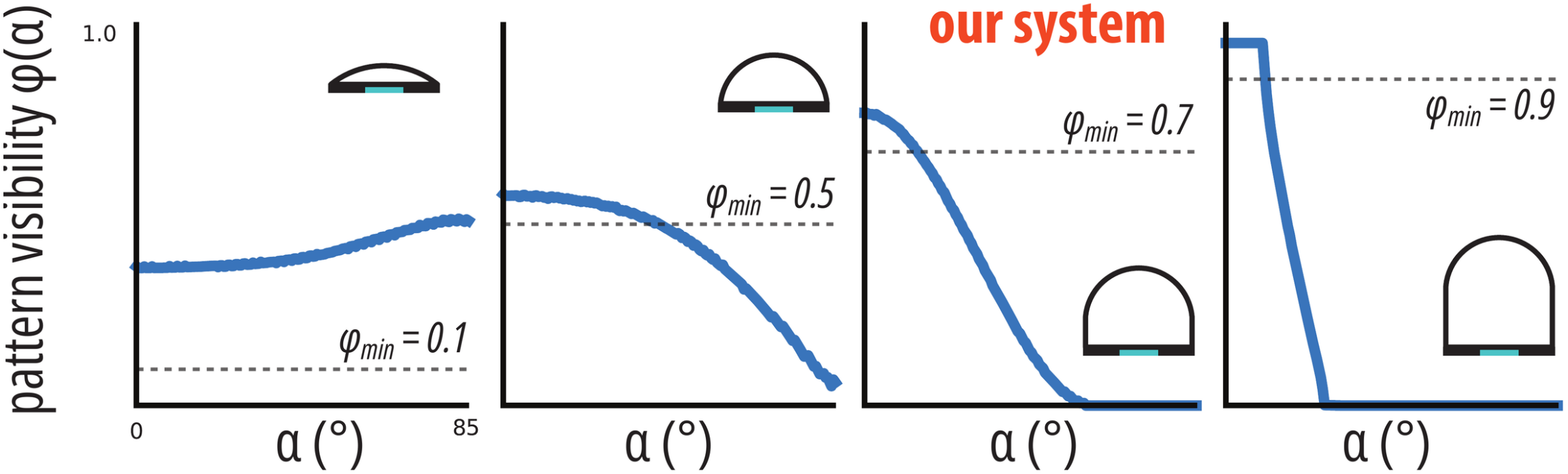
# Pattern Visibility



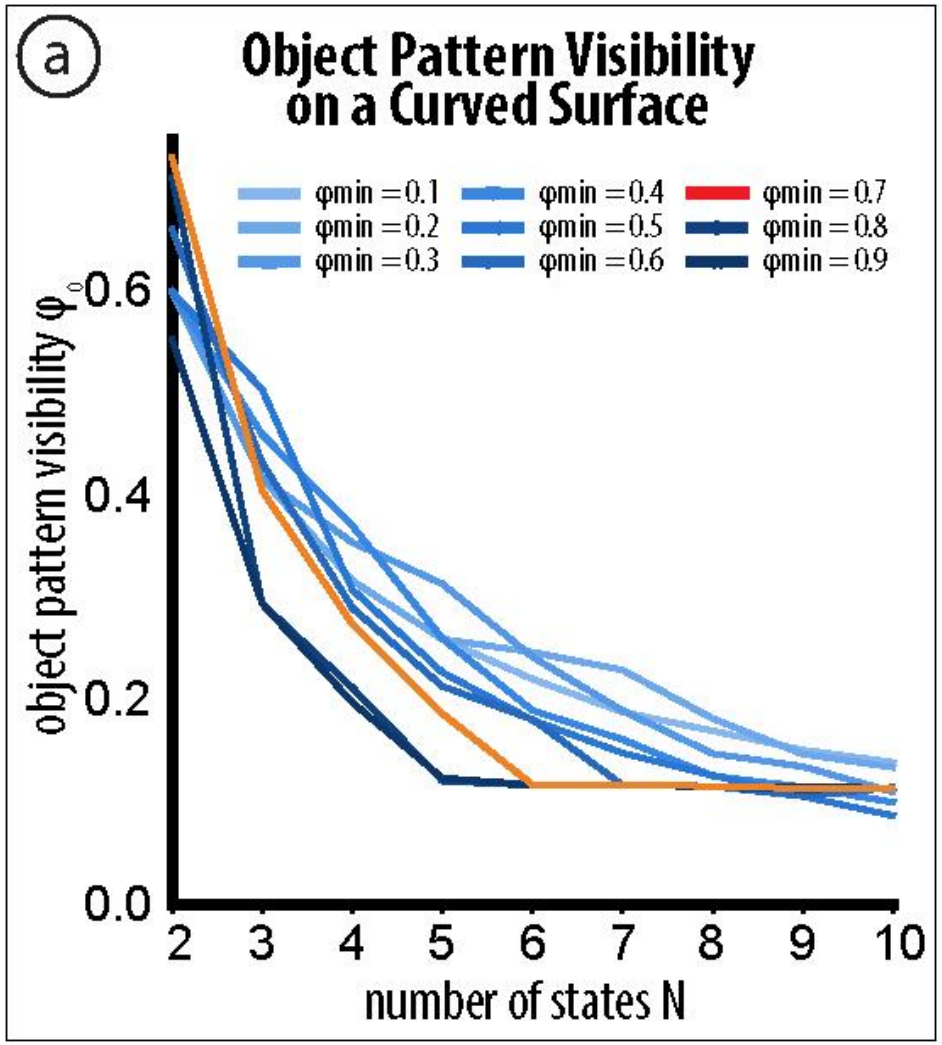


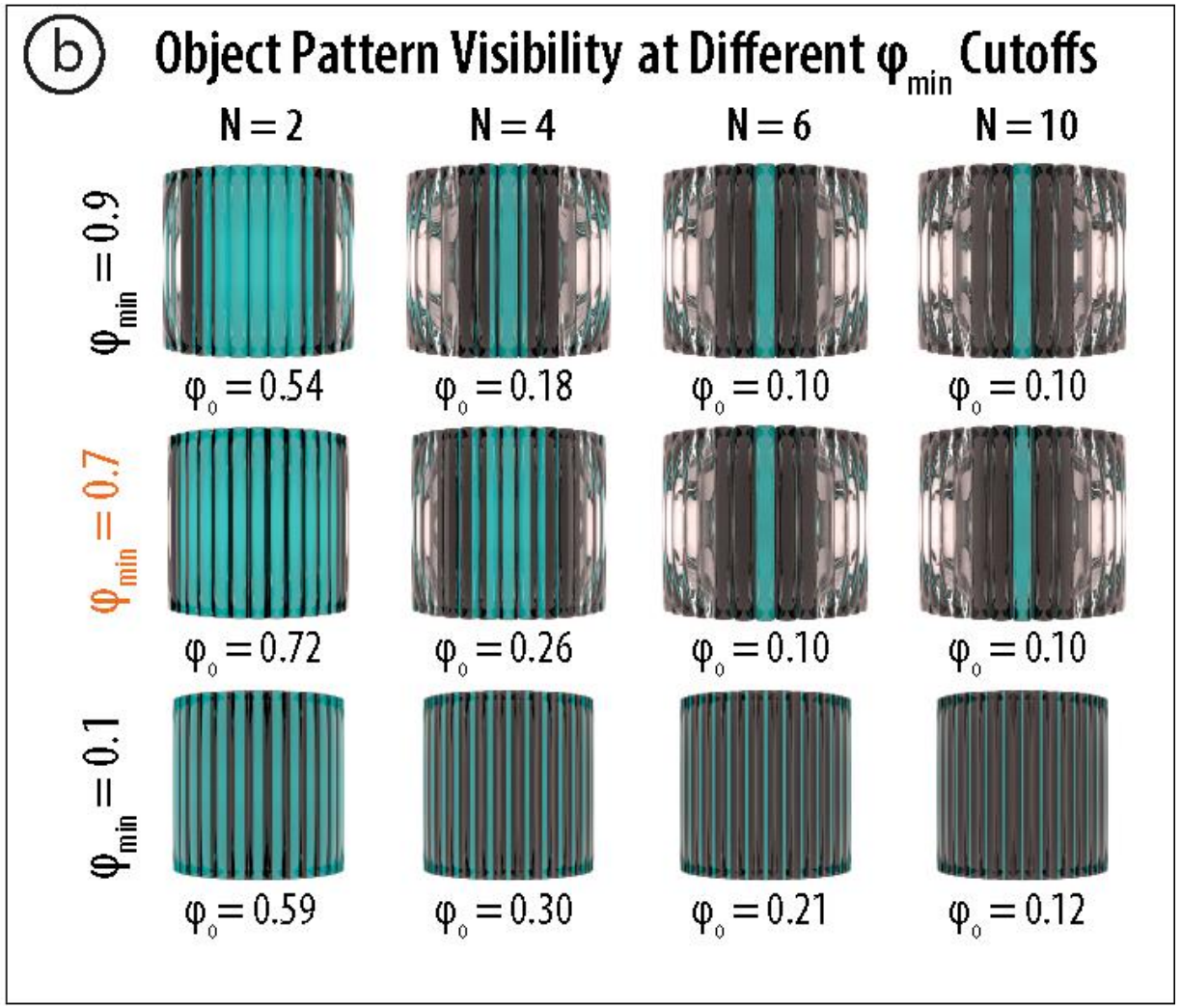

| 0° | 10° | 20° | 30° |
|---|---|---|---|
| φ = 1 | φ = 0.88 | φ = 0.34 | φ = 0 |

# Design Tradeoff

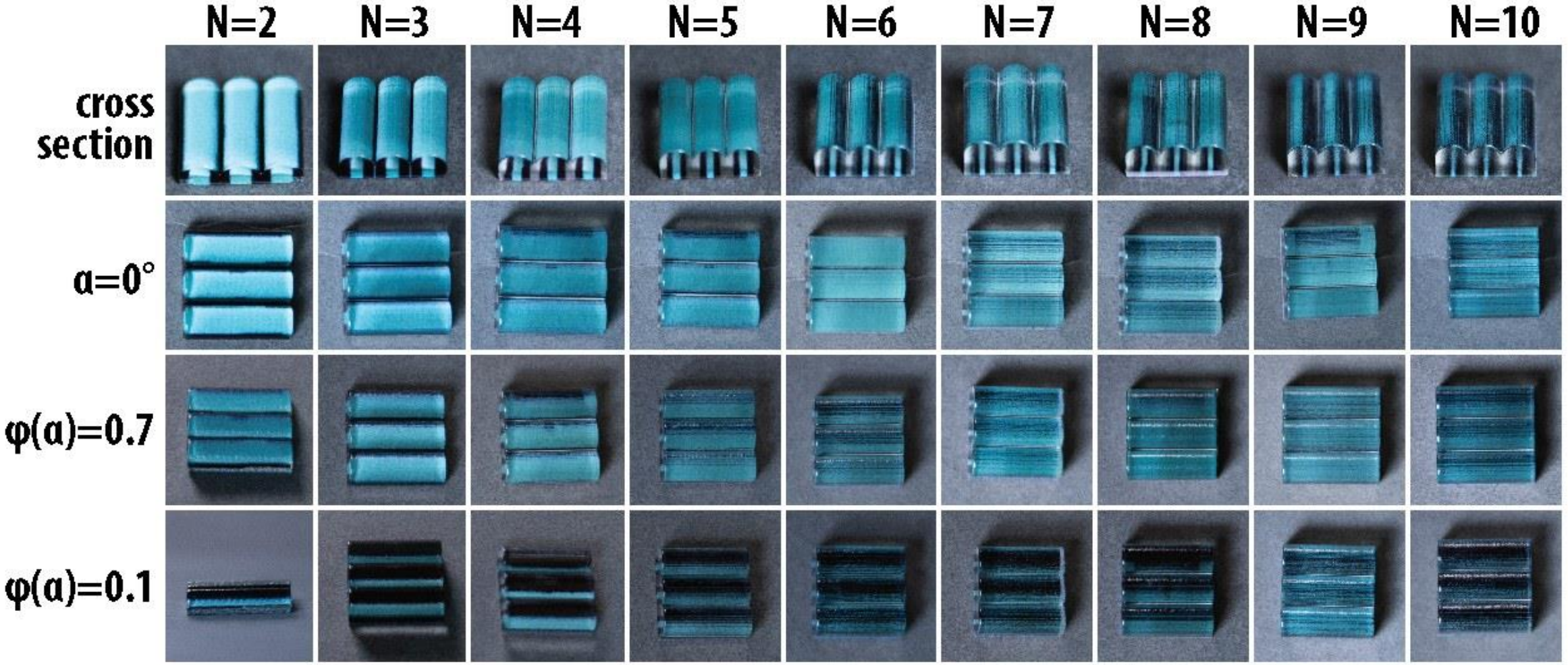


# Lens Optimization

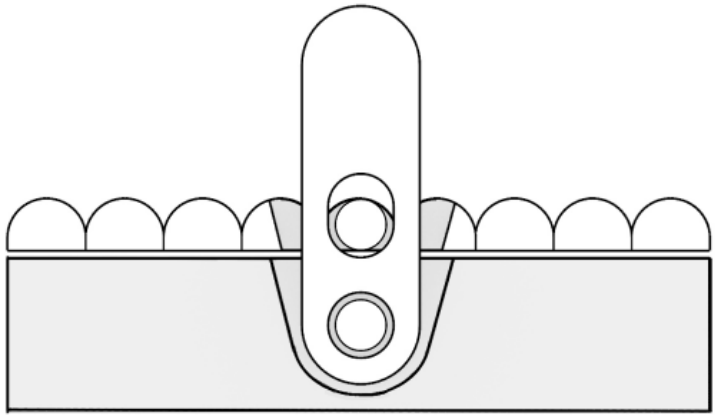


# Physical Validation

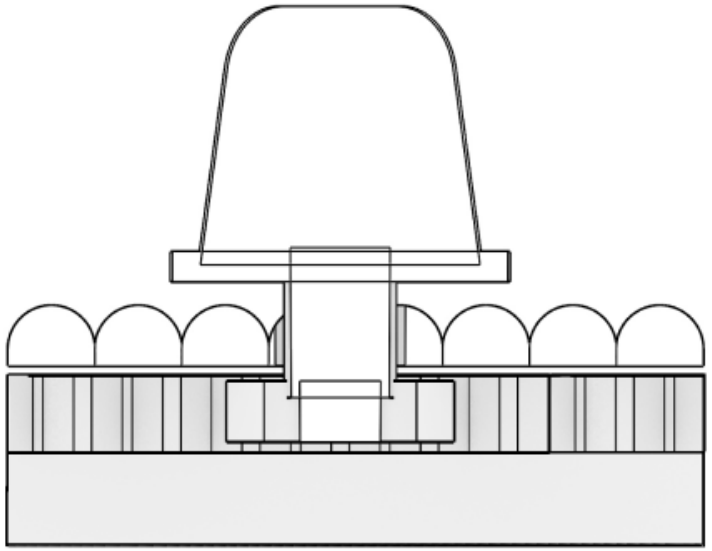


# Actuation Mechanisms



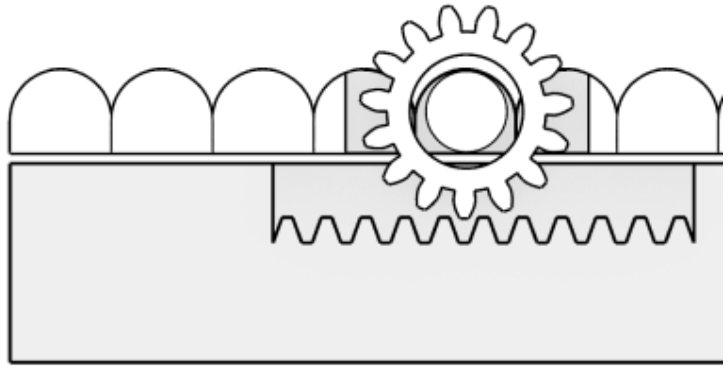
Switch

Two-State



Knob

Multi-State



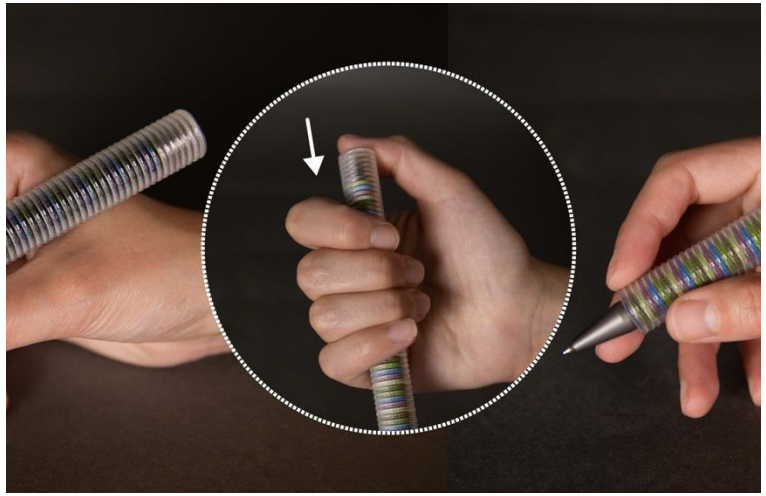
Roller

Smooth

# Native Interaction



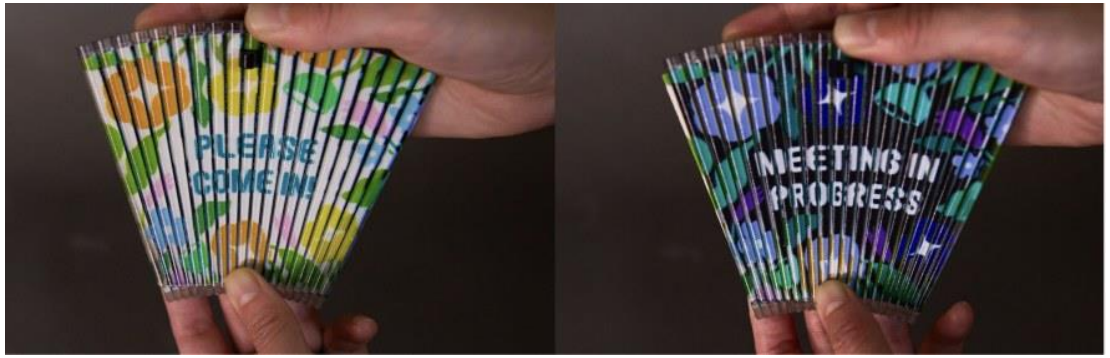




# Fabrication




Simulated




Printed